

Einheit 2 · Winkel berechnen

1) a) Winkel: Umkehrtaste b) $\sin \alpha = \frac{3}{8} = 0,375$, $\alpha = \sin^{-1}(0,375) \approx 22,0^\circ$ c) $\cos \alpha = \frac{5}{7}$, $\alpha = \cos^{-1}\left(\frac{5}{7}\right) \approx 44,4^\circ$

2) a) $\cos \alpha = \frac{460}{610}$, $\alpha \approx 41,1^\circ$ b) $\cos \alpha = \frac{190}{245}$, $\alpha \approx 39,1^\circ$ c) $\cos \alpha = \frac{3}{8}$, $\alpha \approx 68,0^\circ$, also rund 68° d) $\cos \alpha = \frac{7}{15}$, $\alpha \approx 62,2^\circ$, also rund 62° e) $\cos \varepsilon = \frac{9}{26} \approx 0,346$, $\varepsilon \approx 69,7^\circ$, also rund 70° f) $\cos \varepsilon = \frac{7}{18}$, $\varepsilon \approx 67,1^\circ$ g) Teildreieck mit der Höhe: $\sin \alpha = \frac{5,8}{9,6}$, $\alpha \approx 37,2^\circ$ h) $\sin \alpha = \frac{3,9}{8,5}$, $\alpha \approx 27,3^\circ$ i) $x = \sqrt{240^2 + 18^2} \approx 240,7$ cm; $\tan \alpha = \frac{18}{240}$, $\alpha \approx 4,3^\circ$ j) $x = \sqrt{300^2 + 25^2} \approx 301,0$ cm; $\tan \alpha = \frac{25}{300}$, $\alpha \approx 4,8^\circ$ k) $AB = 4$, $AC = 5$; $BC = \sqrt{4^2 + 5^2} \approx 6,40$; $\tan \beta = \frac{5}{4}$, $\beta \approx 51,3^\circ$ l) $AB = 3$, $AC = 7$; $BC = \sqrt{3^2 + 7^2} \approx 7,62$; $\tan \beta = \frac{7}{3}$, $\beta \approx 66,8^\circ$